3 Yr. Degree/4 Yr. Honours 5th Semester Examination, 2025 (CCFUP)

Subject : Computer Science

Course : COMP5013 (MAJOR)

(Software Engineering)

Time: 3 Hours **Full Marks: 60**

The figures in the margin indicate full marks.
Candidates are required to give their answers in their own words as far as practicable.

1. Answer *any ten* questions: 2×10=20

(a) What is software engineering?

(b) What is Alpha testing?

(c) What do you mean by debugging of a software system?

(d) What is use case diagram?

(e) What do you mean by agile software development?

(f) Differentiate between physical and logical DFD.

(g) Mention some software analysis and design tools.

(h) What do you mean by economical feasibility of a software system?

(i) What is system software?

(j) What do you mean by Lines of Code (LOC)?

(k) What is data dictionary?

(l) Define software reliability.

(m) Differentiate between software verification and validation techniques.

(n) What is adaptive maintenance?

(o) How can you find the size of a software product?

2. Answer *any four* questions: 5×4=20

(a) What does the Control Flow Graph (CFG) of a program represent? Draw the CFG of the following program. 2+3

```
        int gcd (int x, int y){
while (x ! = y){
        if (x > y)
        x = x - y,
        else y = y - x;
}printf ("%d", x);
}
```

(b) Discuss the major advantages of the Object Oriented Design (OOD) methodologies over the process oriented design techniques. What do you mean by feasibility report? 4+1

(c) What is SRS document? Write down the characteristics of a good SRS document. 1+4

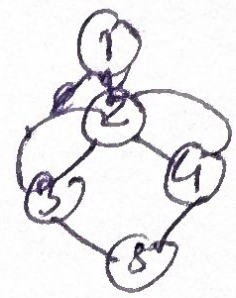
(d) Discuss Black Box testing mechanism. "System testing can be considered as a pure black box test". — Do you agree with the following statement? 3+2

(e) What according to you is a quality software product? Write down any four responsibility of a project manager. 3+2

(f) Write short notes on (*any one*): 5

(i) Rayleigh curve

(ii) Software Reverse Engineering



3. Answer *any two* questions: 10×2=20

(a) Mention the major activities that are undertaken during the development of a software product. What problems might a software development organisation face if it is not following any SDLC for development of a large size software. 8+2

(b) What are the different catagories of software development projects according to the COCOMO estimation model? Give an example of software product development projects belonging to each of these catagories. What is basic COCOMO? 3+2+5

(c) What do you mean by functional independence of a software model? Discuss different types of cohesions that may exists in software module. 3+7

(d) Draw a labelled DFD for Hospital Management System (take your own assumption) up to level 2.